Junxiang Wang

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PROFESSIONAL SUMMARY

Computational mechanics and HPC engineer with strong experience in parallel C++ finite element solvers using MPI on Linux-based HPC systems. Skilled in CMake build systems, modular code design, and large-scale multiphysics simulations for fracture and porous media applications.

TECHNICAL SKILLS

- Numerical Methods: FEM, phase-field fracture, constitutive modeling, multiphysics coupling
- Programming: C++, Fortran, Python, MATLAB
- Simulation Tools: Abaqus (UMAT, UEL), deal.II, Rhino, Grasshopper
- Platforms: Linux, Git, HPC environments
- Experimental: Dynamic triaxial testing, GDS systems, data processing

Certificate |

- National Computer Rank Examination Certification, C-Programming (level 2, highest for non C.S. student).
- National College English Test Certification. (level 6, highest for non English student)

Language |

- English: Fluent
- Italian: Daily Communication
- Chinese: Mother tongue

PROFESSIONAL EXPERIENCE

Politecnico di Milano, Italy | *Post Doc* July. 2025 - Current

Project: dealiiX: an Exascale Framework for Digital Twins of the Human Body

- Incorporating **Phase-Field** factor into the partial differential equations for describing biphasic media fracturing
- Developing **FEM solver** in **C++** and **Fortran** programming language over **Linux** platform
- The **first** contribution provides a theoretical basis for the occurrence of NHF

Shanghai Jiao Tong University, China | *Post Doc* June. 2024 - current

Project: Phase-Field Modeling of Interfacial Damage Coupled with Reaction-Diffusion Behavior

University of Stuttgart, Germany | *Ph.D. Student* Apr. 2022 - Jun. 2023

Project: Mathematical Modelling of Sandstone Natural Hydraulic Fracturing

- Incorporating **Phase-Field** factor into the partial differential equations for describing biphasic media fracturing
- Developing **FEM solver** in **C++** and **Fortran** programming language over **Linux** platform
- The **first** contribution provides a theoretical basis for the occurrence of NHF

University of Padua, Italy | *Ph.D. Student* Oct. 2019 - Apr. 2022

Project: Mathematical Modelling of soil subjected to cyclic loading

- Comprehensive review of porous media modelling, computational mechanics and non linear mechanics
- A cyclic constitutive model that could achieve the typical deformation characters of soil under cyclic(dynamic) loading such as: shakedown, degradation and strain accumulation is developed
- An **ABAQUS UEL-UMAT** code was built to realize the constitutive model

HuBei University of Technology, China | *Master Student* Sep. 2016 - Jun. 2019

Project: Research on Dynamic Characteristics of Expansive Soil

- Conducting the dynamic triaxial experiments with different confining pressure and loading frequency
- Analysis the deformation properties such as hysterics, stiffness degradation of expansive clay under different dynamic loading conditions
- Using **python** to complement the process of data analyzing

Deutsch—Chinesischer Kulturaustausch fur Kunst und Design e.V., Germany | *Exchange Student* Apr. 2017 - Oct. 2017

Project: International Cooperation Project about “One Belt and One Road Initiative”

- Designing exhibition hall by using Rhino and Grasshopper with the local professor and students
- Using **Python** and **Abaqus** to perform the structural analysis
- Working as a translator for the delegation from China

EDUCATION

Ph.D. in Structural and Solid Mechanics Oct. 2019 - Jun. 2023

University of Padova, Padova, Italy

Joint Ph.D. Course in Structural and Solid Mechanics Apr. 2022 - July 2022

University of Stuttgart, Stuttgart, Germany

Bachelor and Master Degree in Civil Engineering Sep. 2012 - Jun. 2019

Hubei University of Technology, Hubei, China

SELECTED PUBLICATIONS

1. Liu J., **Wang J.**, and Wang L. (2025). A nonlocal mixed-mode fatigue crack growth model based on peridynamic differential operator theory, *Computer Methods in Applied Mechanics and Engineering*, (Accepted).
1. **Wang J.**, Sonntag A., Lee D., Xotta G., Salomoni V., Steeb H., Wagner A. and Ehlers W. (2024). Modelling and simulation of natural hydraulic fracturing applied to experiments on natural sandstone cores, *Acta Geotechnica*, 19(11), 7709-7725.
2. **Wang J.**, Xotta, G., De Marchi, N., and Salomoni, V. (2022). An Enhanced Bounding Surface Model for Modelling Various Cyclic Behaviour of Clay. *Materials*, 15(21), 7609.
3. Zhuang X, **Wang J**, Wang K, Li K, Hu Z (2018). Experimental study on dynamic characteristics of expansive soil modified by weathered sand. *Rock and Soil Mechanics, Chinese Academy of Sciences*. 39(S2): 149-156.
4. Zhuang X, **Wang J**, Li K, Wang K, Hu Z (2019). Comparative study on characteristic of hysteretic curves of expansive soil improved by weathered sand. *Chinese Journal of Rock Mechanics and Engineering*. 38(s2):3709-3716.

CONFERENCE PRESENTATION

- Wang, J., et al. "Modelling of natural hydraulic fracturing in sandstone cylinders" *The Seventh International Conference on Computational Modeling of Fracture and Failure of Materials and Structures. 2023*.
- Wang, J., et al. "Implementation of a bounding surface constitutive model for fully coupled dynamic analysis of soil and its validation using dynamic triaxial test." *8th European Congress on Computational Methods in Applied Sciences and Engineering-ECCOMAS Congress 2022.*
- Wang, J., et al. "Anisotropic Bounding Surface Plasticity Model for Porous Media Subjected to Complex Loading Condition." *CONVEGNO AIMETA. 2022.*

SPECIALIZED TRAINING

Numerical Method and Modelling | *University of Stuttgart, Germany* Apr. 2022

- Course of the Theory of Porous Media

Computational Plasticity | *Universitat Politecnica de Catalunya, Spain* Sep. 2021

- Short Course on Computational Plasticity

Constitutive Modelling | *Dassault Systèmes, Italy* May. 2021

- Writing User Subroutines with Abaqus

CERTIFICATIONS & LANGUAGES

University of Padova, Italy | *Full International scholarship* Jun. 2019

University of Stuttgart, Germany | *Full International scholarship* Apr. 2022

Hubei University of Technology, China | *Excellent College Student* 2013 - 2015